

* OSI Model

Layer 7 - Application Layer

- Provides network services to applications
- Interface between user & network

> Protocols:

- HTTP/HTTPS • FTP • SMTP/POP3/IMAP • DNS • Telnet / SSH

> Cyber Attacks:

- SQL Injection • XSS • CSRF • Authentication attacks

> Tools:

- Burp Suite • OWASP ZAP

Exam line → Application Layer provides services to user applications.

Layer 6 - Presentation Layer

Translator + Security Layer

- Encryption/Decryption
- Compression
- Format Conversion

> Ex: • SSL/TLS (HTTPS) • JPEG, MP3 encoding

> Cyber Attacks:

- SSL Stripping
- Weak cipher exploitation

→ Encryption happens at Presentation Layer.

Layer 5 - Session Layer

Session Manager

- Establishes, maintains, terminates sessions
- Controls dialog (half, full duplex)

Ex: Login session, video call sessions

> Cyber Attacks:

- Session hijacking
- Session fixation

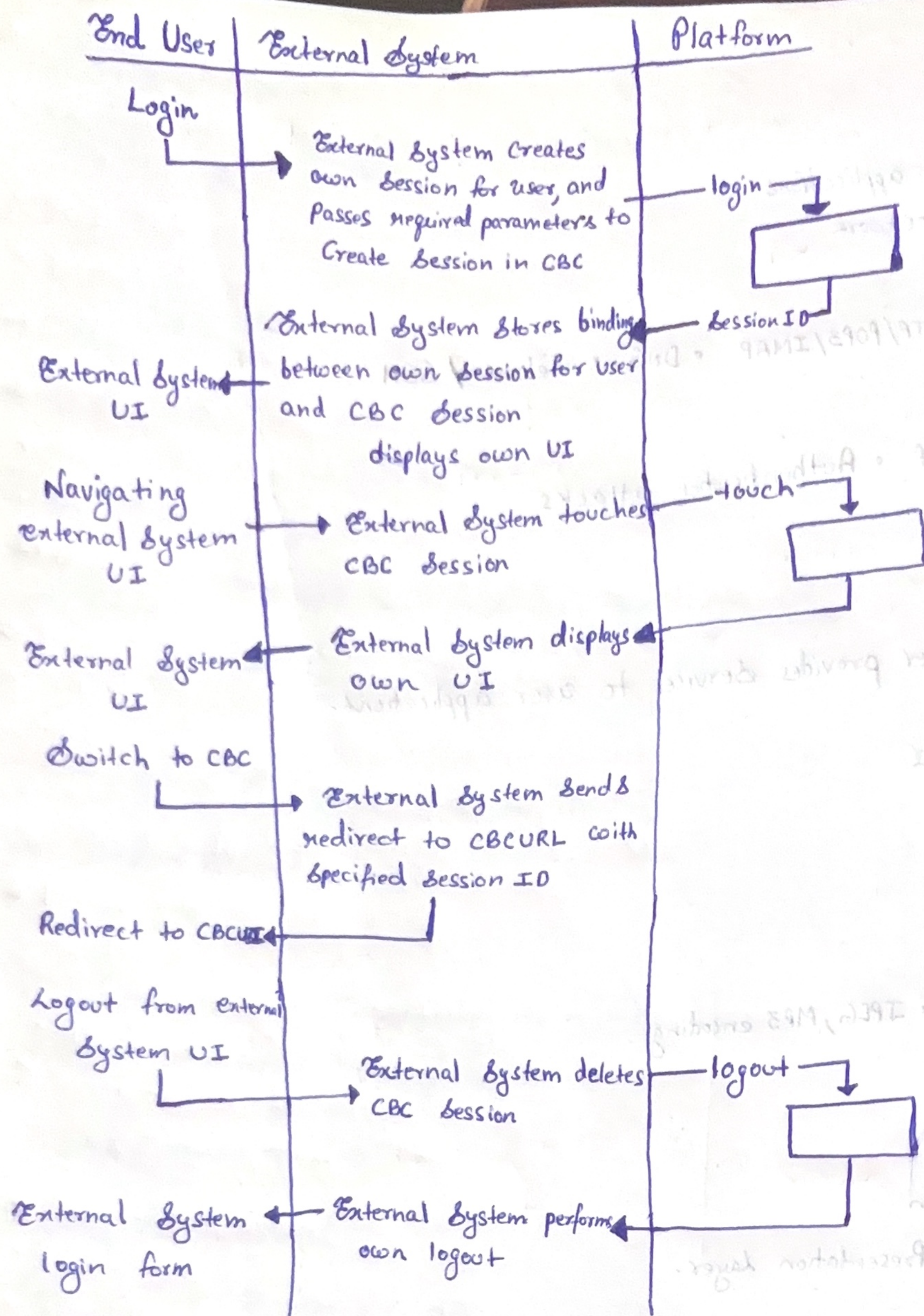


Diagram Structure

The diagram is divided into three vertical swimlanes. The swimlanes diagram showing session integration between an External System and a platform (CBC) from the End user perspective.

1. End User - what the user sees and does
2. External System - logic + session handling
3. Platform (CBC) - backend platform session lifecycle.

-1. End User Lane (user Action & UI)

This lane shows user-visible actions

Step-by-step:

- Login
- user login into the External System

- ◆ Authentication happen only here, not on CBC

Important - User never directly logs into CBC initially.

- External System UI

- ◆ After login, the user interacts with the External System UI
- ◆ All navigation is still within the external system.

- Navigation External System UI

- ◆ user continues normal actions
- ◆ Meanwhile, background session work happen (in next lane)

- Switch to CBC

- ◆ Go to CBC
- ◆ open platform

→ This is a content switch, not fresh login

- Redirect to CBC UI

- ◆ Browser is redirect automatically
- ◆ user lands inside CBC already authenticated

- Logout from External System UI

- ◆ users logout from external system

- External System login form

- ◆ User is fully logged out
- ◆ Login Screen appears again

2- External System Lane (Most Important Logic)

→ This lane controls session orchestration

- ◆ External System Creates own session

- After user login:

- External System creates it's own session
- Generates session parameters

- These parameters are required to create a CBC session

→ Security Concepts:

- Session abstraction
- External System act as identity broker

- ◆ Stores binding between sessions

- External System stores a mapping:

Sql
↳ External Session ID ↔ CBC Session ID

- The binding is critical for:

- Session tracking
- Logout Synchronization

◆ Touches CBC Sessions

- External System sends a request to CBC:
 - "User is active"
- Keep CBC Session alive
- Prevent timeout

→ This is called : Session Keep-alive / heartbeat

◆ Displays own UI

- Even while CBC Session exists
- User still see External System UI

→ CBC Session is background-only at this stage

◆ Redirect to CBC URL (Sends)

- When user switches:
 - External System sends browser redirect
 - Redirect Contains CBC URL + Session ID
- Ex: `https://cbc.platform/app?sessionID=XYZ`

◆ Deletes CBC Session

- on logout:
 - External System explicitly deletes CBC Session
- Prevents:
 - orphan Session
 - Security leaks

◆ Performs own logout

- Clears it's own Session
- User is fully logout everywhere

3- Platform (CBC) Lane (Session Life Cycle)

This lane shows CBC's backend reactions

◆ Login

- CBC receives Session Creation request
- Creates CBC Session
- Returns Session ID

→ CBC Trusts the external System.

◆ Session ID

- CBC Session Identifier
- Used for:
 - Redirect login
 - Session validation

◆ Touch

- CBC receives keep-alive calls
- Extends Session validity

- ◆ Logout
- CBC session is destroyed
- User access is revoked immediately

Layer - 4 - Transport Layer

End-to-End delivery

- Segmentation & reassembly
- Flow Control
- Error Control

Protocols:

TCP

- Reliable
- 3-way handshake
- HTTP, FTP

UDP

fast

No handshake

DNS, VoIP

> Cyber Attacks:

- Syn Flood
- UDP Flood
- Port Scanning

Important → Port exist at transport layer

Layer - 3 - Network Layer

Routing + Logical addressing

- Logical addressing (IP)
- Packet forwarding
- Path Selection

Protocols:

- IP
- IPsec
- ICMP
- Routing Protocols

> Routing Protocols (Important Addition)

◆ RIP (Routing Information Protocol)

- Distance - vector
- Metric: Hop count
- Max hop: 15
- old & insecure

→ Attack: Routing Poisoning

◆ OSPF (Open shortest path first)

- Interior Gateway Protocol (IGP)
- Link - state
- Metric: cost

• user areas (Area 0 backbone)

> Attacks: • fake LSA injection • OSPF flood attacks

• BGP (Border Gateway Protocol)

• Exterior Gateway Protocol (EGP)

• Internet backbone protocol

• Path-vector

• Uses AS numbers

> Attacks:

• BGP Hijacking • Internet-wide outages

• Prefix leaks

→ Important: Routing protocol operates at Network Layer

Layer-2- Data Link Layer

MAC addressing & frames

• MAC addressing

• framing

• Error detection (CRC)

Protocols:

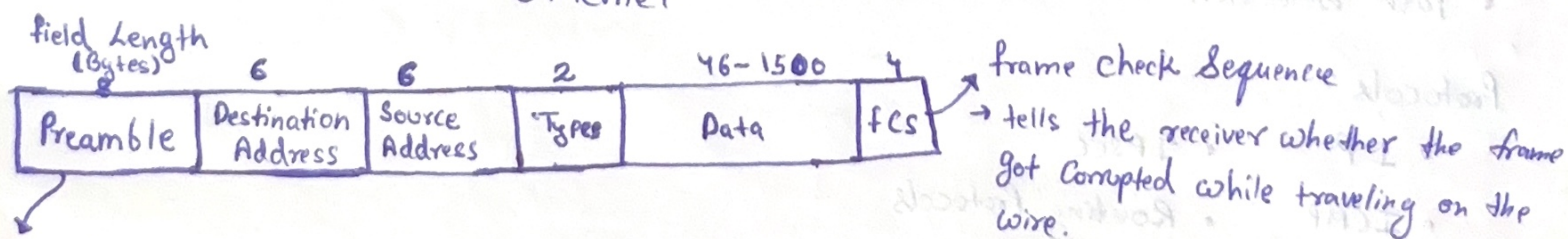
• ARP • Ethernet • VLAN

> Cyber Attacks

• ARP poisoning • MAC flooding • MITM (LAN)

Cyber fact: Most LAN attacks happens at Layer 2

Ethernet



A fixed bit pattern sent before every ethernet frame to help the receiver synchronize with the sender

In simple words: "Hey wakeup, data is coming - get sync"

It is not actual data, not IP, not MAC - it's a timing & synchronization signal

What happens without a ~~pre~~ preamble?

→ If preamble didn't exist:

• Receiver may:

• Miss first bits

• Misread MAC address

• Corrupt entire frame

• Result:

CRC error (Cyclic Redundancy check)
frame dropped

Layer-1 - Physical Layer

Bits & Signals

- Transmits raw bits (0s & 1s)

- Hardware & Signals

Ex: • Ethernet cable, fiber optics, wifi radio waves

Attacks:

~~wire~~ • wire tapping • Signal Jamming • Hardware damage

* Data Flow (Encapsulation)

Sender:

L7 → L1 (adds headers)

Receivers:

L1 → L7 (removes headers)